RESEARCH PROPOSAL

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Designing Digital Systems software for Kidney Patient's monthly medical reports Using UI/UX Principles.

1. Introduction

Most of the time, kidney patients experience large delays and hardships in obtaining their medical reports since this would necessitate personal visits to providers or waits for paper-based reports. These pose barriers to timely interventions and could lead to deterioration in health outcomes. Where digital platforms present an opportunity to solve these problems by instantly having access to medical information, many current systems are not tailor-designed for the unique needs of kidney patients; hence, the consequence is often an experience that is poorly usable, frustrating, or even leading to misunderstandings of critical health data because of complex interfaces, lack of accessibility features, or overwhelming medical jargon.

Considering the patients with kidney diseases may have physical limitations, such as movement disabilities, and cognitive problems, like not being able to remember or concentrate, a friendly and accessible digital system is necessary. It must be characterized by clarity and intuitiveness of navigation, with lettering at least big in size, no need for scrolling, and simple menus where one can find all necessary information without wasting too much time. Accessibility features, such as voice-assisted navigation, variable font size, and high-contrast modes, are critical in ensuring that patients with visual or motor impairments can also easily use the system.

Therefore, this study shall focus on designing a digital system for patients with kidney issues that incorporates usability, accessibility, and high security of data, with the view of ensuring timely, secure exposure to their clinic reports for their proper involvement in their health management and with minimal irritations usually associated with traditional or poorly designed digital platforms.

Literature Review

1. The most critical findings of the research were on the design and usability testing of a personal health record for patients with CKD. Twelve usability issues were identified from the analysis, including three related to user control, three with consistency and standards, one guiding users to diagnosis, and five with error prevention. These findings indicate the need to ensure that the clinical standards are observed and the system is user-friendly because previous failures in similar systems were partly due to issues of non-compliance with usage and expectation by users. Participation of experts in health information management and medical informatics during the developmental stages further ensured that this software was appropriate both technically and within the healthcare context. Ultimately, the study testifies to how compliance and usability go hand in hand if personally effective health record systems that enhance patient care are to be further improved.

Salehi, F., Habibi, M. R. M., Sharifian, R., Kazemzadeh, F., & Setoodefar, M. (2024). Empowering chronic kidney disease patients: A personalized electronic health record system for enhanced care and management. *Frontiers in Health Informatics*, *13*, 194.

2. The study found that the Renal ePROM system-electronically adapting two PRO questionnaires-was, in general, well accepted by patients with advanced CKD. Patients in this study (mean age 64.3 years) thought the system was easy to use: the average score from questions assessing usability and satisfaction was 4.6 out of a possible 5. The average completion time was 15.9 minutes for the two questionnaires combined. However, less-experienced users of electronic devices and the Internet had more difficulties. In general, this study appears to indicate that the Renal ePROM system can be a useful instrument complementing clinician-reported outcomes and supporting the care of patients with advanced CKD.

Aiyegbusi, O. L., Kyte, D., Cockwell, P., Marshall, T., Dutton, M., Walmsley-Allen, N., ... & Calvert, M. (2018). Development and usability testing of an electronic patient-reported outcome measure (ePROM) system for patients with advanced chronic kidney disease. *computers in biology and medicine*, *101*, 120-127.

3. Renal failure is the internal problem that most people do not pay much attention to. This is a severe health condition, but in most cases, it goes unnoticed since one may not easily realize it

without the intervention of a medical professional. To curb that challenge, an e-Renal Failure Diagnosis System has been devised, which helps an individual in making diagnoses of renal failure remotely. This system consists of two major modules: a diagnosis module, through which users can perform self-diagnosis, and an information module that contains information related to renal failure, including a directory of dialysis centers within Peninsular Malaysia. The system is developed using the expert system methodology with a prototyping model, production rules for knowledge representation, and pattern matching for diagnosis searches. It aims to care for society by providing access to the assessment of kidney health, simulating expert medical knowledge for the users.

Noh, N. M., Zakaria, Z., & Kasim, S. (2019). Online Diagnose System for Risk of Kidney Failure. *International Journal of Advanced Science Computing and Engineering*, *1*(1), 48-67.

4. Digital health is emerging as an important tool in the quest to improve patient care, especially for those suffering from chronic conditions such as CKD, where dietary self-management presents significant challenges. It seems obvious that digital health interventions increase access to information on nutrition and dietitians more frequently, with the possibility of frequent monitoring and feedback on dietary management in CKD patients. Although early evidence looks promising, further research is needed to complete the evaluation of how these interventions can be integrated into routine care. Other design and implementation considerations for such digital interventions are thorough consideration of the needs, taking into consideration also the digital literacy of the patients; developing an intervention which would fit well into existing nutrition services; and being able to embed this tool seamlessly within the clinical care process. The purpose of this review is to outline the current state of literature on digital health for self-management in CKD, debate issues that should be taken into account when implementing digital interventions, and give practical lessons useful to translate into the development of digital tools to improve nutritional care for CKD patients.

Dawson, J., Lambert, K., Campbell, K. L., & Kelly, J. T. (2024, July). Incorporating digital platforms into nutritional care in chronic kidney disease. In *Seminars in Dialysis* (Vol. 37, No. 4, pp. 350-359).

5. CKD is a worldwide health burden that severely impairs the quality of life of patients and increases demands on healthcare. More recently, digital health symptom management interventions have emerged as possible solutions for such patients. However, if the knowledge is to be used effectively in a clinical setting, an understanding is needed of their development, validation, and effectiveness. In this scoping review, the existing literature related to digital health interventions for symptom management for people with CKD was mapped using the UK Medical Research Council's complex intervention framework.

This review aimed to analyze and synthesize results from 31 studies from mainly developed countries, which showed the potential of digital interventions in improving symptom management, quality of life, and engagement by patients. However, several gaps were found: a lack of iteration refinement that involves multidisciplinary stakeholders; underutilization of theory-driven and evidence-based approaches; and a limited focus on long-term implementation and process evaluation. These findings suggest that while digital health interventions are promising, studies are further needed to address these challenges and optimize such technologies in integrating routine CKD care. Future studies now need to emphasize patient-centered design, comprehensive validation, and the exploration of long-term outcomes to ensure effectiveness in clinical practice.

ZHENG, X., Zhen, Y. A. N. G., Shu, L. I. U., Yuqian, L. I., & Aiping, W. A. N. G. (2024). Toward the complexities of the development and validation process of digital health interventions for the symptom management for patients with Chronic Kidney Disease: A scoping review based on the UK Medical Research Council Framework.

Research Problem

Broad Topic:
Designing Digital Systems software for Kidney Patient's monthly Medical reports Using UI/UX Principles.

Narrowed Topic
Developing a user-friendly and secure digital platform tailored to kidney patients for timely access to medical reports and health management.

Research Problem Statement
Kidney patients often experience significant delays and challenges in accessing medical reports and clinical data through traditional methods, hindering their ability to manage their condition effectively. Existing digital solutions are not adequately designed for the specific cognitive and physical limitations of Chronic Kidney Disease (CKD) patients, leading to usability issues and potential misinterpretation of critical health information. Furthermore, ensuring the security and privacy of sensitive health data remains a significant concern. This study seeks to address these gaps by developing a digital health platform that combines ease of use, accessibility, and data security to improve the management and health outcomes of kidney patients.

Objectives

The development of digital systems for the access of clinic reports online by kidney patients is, while overall a very valuable solution, also a unique challenge in and of itself. The thing is, patients with kidney failure are usually cognitively and physically limited; hence, designing easy-to-use, accessible interfaces is basically the need of the hour. There is a keen need to embed principles of UI and UX design in this regard to reduce cognitive loads. This would involve clear navigation, larger icons, and readable fonts with minimal scrolling. The approach would be mobile first since most the patients would prefer smartphones to manage any health need.

Continuous monitoring in kidney disease management is very important, and all patients have to monitor health parameters like blood pressure, GFR, and creatinine levels. Automated notifications will help the health professional when every measure has moved out of the normal range for immediate interventions. The system should further facilitate telehealth consultations with a nephrologist so that patients can do virtual appointments, thus reducing the need for frequently visiting nephrologists in person.

Timely access to medical reports is critical. Electronic systems can permit immediate updates, allowing lab results, prescriptions, or dialysis schedules to be made available immediately. This will definitely empower patients toward active participation in caring for themselves and ensure timely interventions. Other features of the system, such as personalized reminders and the patient's ability to log in every day with his or her health data, further support self-management. In turn, the data will also be available to the doctors to make better informed decisions on treatment. These need to balance each other to create an easy-to-use, accessible, yet data-secure patient-centric system tailored for patients with kidneys.

Significance

Kidney disease requires periodical health parameter monitoring, and the conventional mode of access to medical reports by way of house visits or delayed communication leads to delays in the care. The following study will focus on bridging that gap by developing an amiable and easily accessible digital system that gives timely access to kidney patients regarding their clinic reports and medical data. The system will increase patient engagement through intuitive access to medical information. If patients can easily see and understand their health data, they are more likely to adhere to treatment plans, make smarter decisions, and have improved health outcomes.

The UI/UX design for this system will be targeted specifically for kidney patients, most of whom may have mobility or cognitive challenges. Such a design includes features like clear navigation, large icons, and fonts that are accessible, making the use of the system easier to reduce frustration and misinterpretation of critical health data. What's more, security and privacy will be accordingly implemented to guard sensitive medical information in accordance with healthcare data regulations.

The study contributes to the digital transformation in health by bringing a patient-centered approach to CKD management. This work will directly solve the problem of poor timeliness of access to medical reports and set the path toward future innovation in chronic care. This study strongly underscores user-friendliness, accessibility, and data protection for vulnerable groups of patients, particularly those with CKD, to ensure a safer and more efficient personalized health service.

Methodology

Research Design

This study will adopt a mixed-methods design that combines qualitative and quantitative approaches in the comprehensive investigation into the development, usability, and effectiveness of the proposed digital platform for kidney patients. The methodology will be realized through three phases: (1) System Design and Development, (2) Usability Testing, and (3) System Evaluation.

(1) System Design and Development

Concept Development: Develop initial design concepts and prototypes based on the needs assessment. This will include wireframes and mockups of the digital system.

Needs assessment shall be done through a critical literature review and consultations healthcare IT experts, and CKD patients to understand specific needs and challenges which the kidney patients face in accessing medical reports to manage their conditions.

Requirements Gathering: Gathering information on key features shall be done, including real-time access to medical reports, personalized alerts, ease of usage by cognitively and physically constrained patients, and high-level security for patient information.

System Development: The **Agile methodology** shall be used to develop this system to make sure that there is continuous input from the stakeholders-patients, clinicians, and IT experts-during the development process. The platform shall be built based on two major modules:

Lab Results/ Diagnosis and Health Monitoring Module: Through this module, the patient can view lab results, clinical reports, and treatment plans.

Information and Support Module: The educational resources provided in this module will help patients understand more about kidney disease and dialysis centers. In addition, the system provides personalized alerts on the medicines and dialysis sessions.

(2) Usability Testing

Testing: Conduct thorough usability testing with a larger group of kidney patients to evaluate the system's effectiveness in real-world scenarios. Assess the system's performance in terms of user satisfaction, ease of use, and accuracy of medical report access.

Sample: A purposive sample of 20–30 CKD patients will be recruited for usability testing; the sample should represent diversity in age, gender, and digital literacy.

Procedure: Patients will be asked to perform common activities on the system, such as looking at a lab report and setting medication reminders. Usability will be tested using Nielsen's usability checklist on criteria including ease of system navigation, clarity of flow of information, and accessibility to cognitively and physically impaired patients.

Data Collection: The collection is done through semi-structured interviews to capture feedback with regard to system functionality, ease of use, and areas that need improvement. Quantitative data will be acquired through a satisfaction survey.

(3) System Evaluation

Data Security Evaluation: Ensure the system adheres to healthcare data security and privacy regulations. Perform security audits and tests to confirm that patient data is protected.

Security and Privacy: The security audit will be performed to protect the data and opinions from patients based on perceived data security.

Health Outcomes: Health data pre- and post-intervention, which includes treatment adherence and laboratory test results, are collected to measure the effect of the system on health management by the patients.

Analysis: Analyze the data collected from usability testing and security evaluations. Measure the impact of the digital system on patient engagement, understanding of medical reports, and overall satisfaction.

Qualitative Data: Interview data are analyzed using thematic analysis to identify key themes related to usability and patient experience.

Quantitative data from the survey will be analyzed using descriptive and inferential statistics to assess overall satisfaction, ease of use, and impact on engagement and health outcomes. 6. Ethical Considerations Approval will be sought from the appropriate institutional ethics review board. Informed consent will be sought from all participants, and all patient data will be protect patient privacy.

It forms a comprehensive basis on which to develop, test, and evaluate a digital health platform that is appropriate for the improvement of access to healthcare and its outcomes among people with kidney disease.

Reporting: Compile the findings into a comprehensive report detailing the design process, testing results, and recommendations for future improvements.

Time Frame

Task	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
1.Select the topic									
2.Finalaizing Research proposal									
3. Literature Review									
4.Design Research methodology									
5.Data collection									
6.Data analysis									
7.Review and Revisions									
8.Final Submission									